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Utsav Karia 17

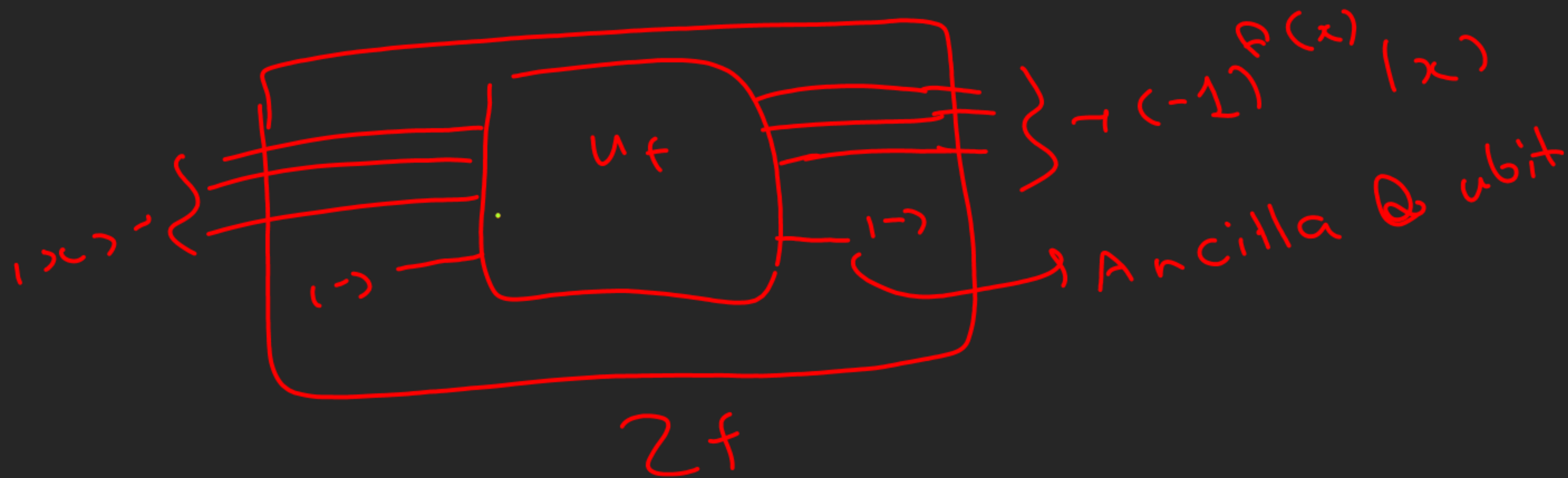
utsavkaria11@gmail.com

utsavkaria-tech → Github → Notes will be shared

→ Phase Query Gates

$$U_f(|x\rangle|a\rangle) = |x\rangle|a \oplus f(x)\rangle$$

$$Z_f|x\rangle = (-1)^{f(x)}|x\rangle$$



$$OR(x) = \begin{cases} 0 & \text{if } x = 0^n \\ 1 & \text{if } x \neq 0^n \end{cases}$$

$$Z_{OR}(x) = \begin{cases} 1|x\rangle & \text{if } x = 0^n \\ -1|x\rangle & \text{if } x \neq 0^n \end{cases}$$

→ Grover's Algorithm

Step-1 → Initialize, set n -qubit to $H^{\otimes n} |0^n\rangle$

Step-2 → Apply Grover Operator t times

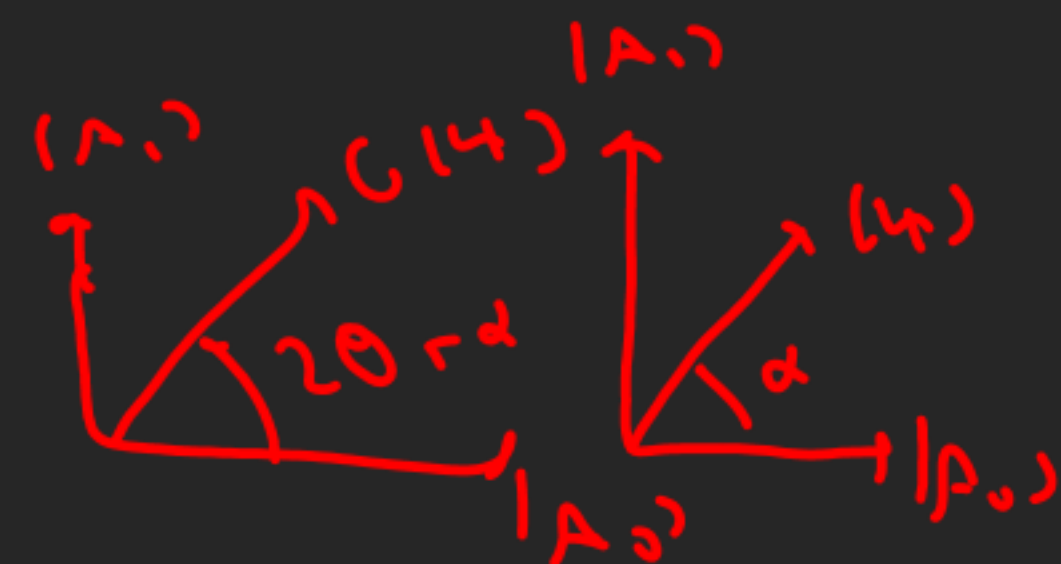
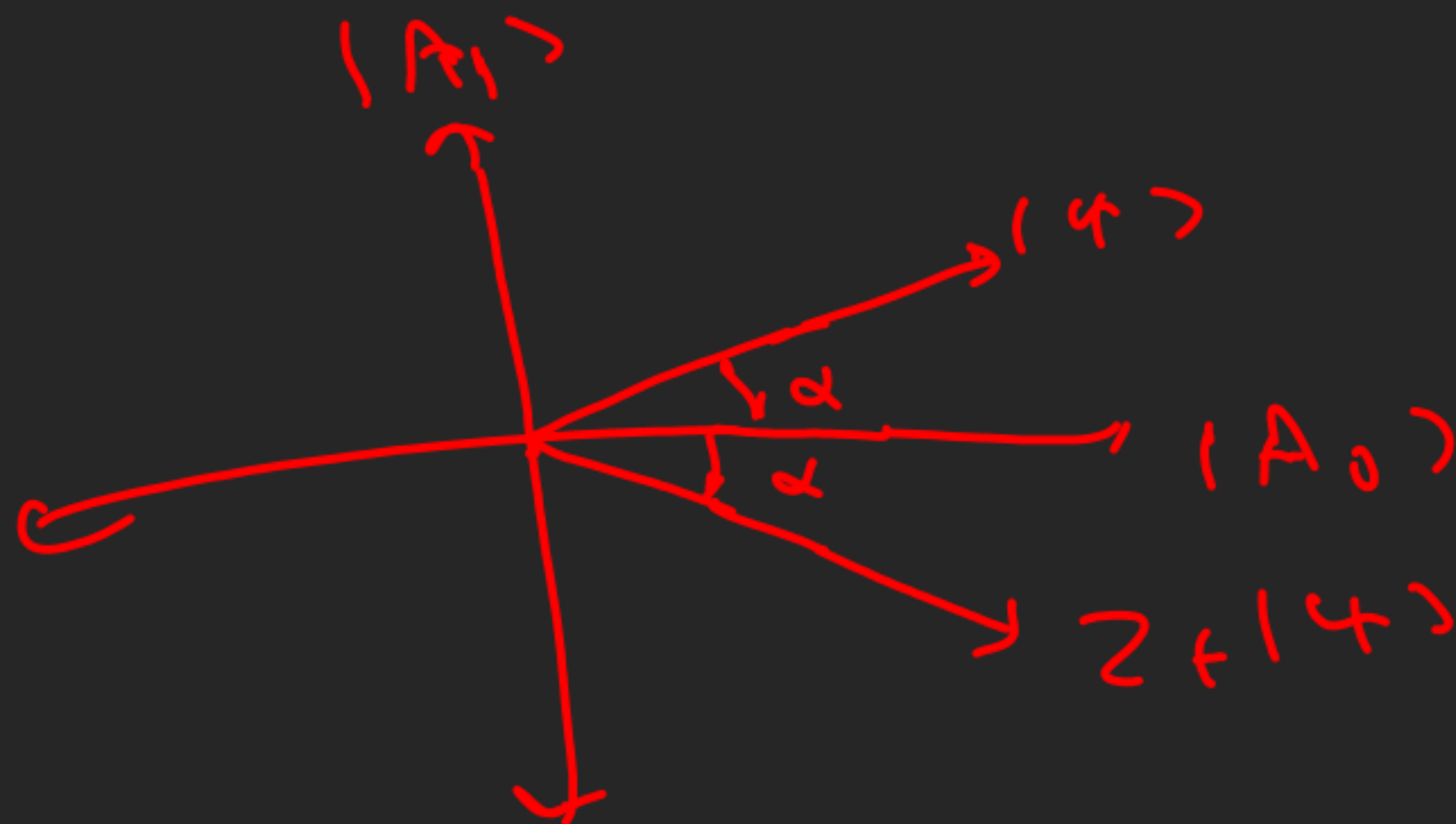
Step-3 → Measure standard basis measurement

Geometric Picture

→ Z_+

$$Z|A_0\rangle = |A_0\rangle$$

$$Z|A_1\rangle = -|A_1\rangle$$

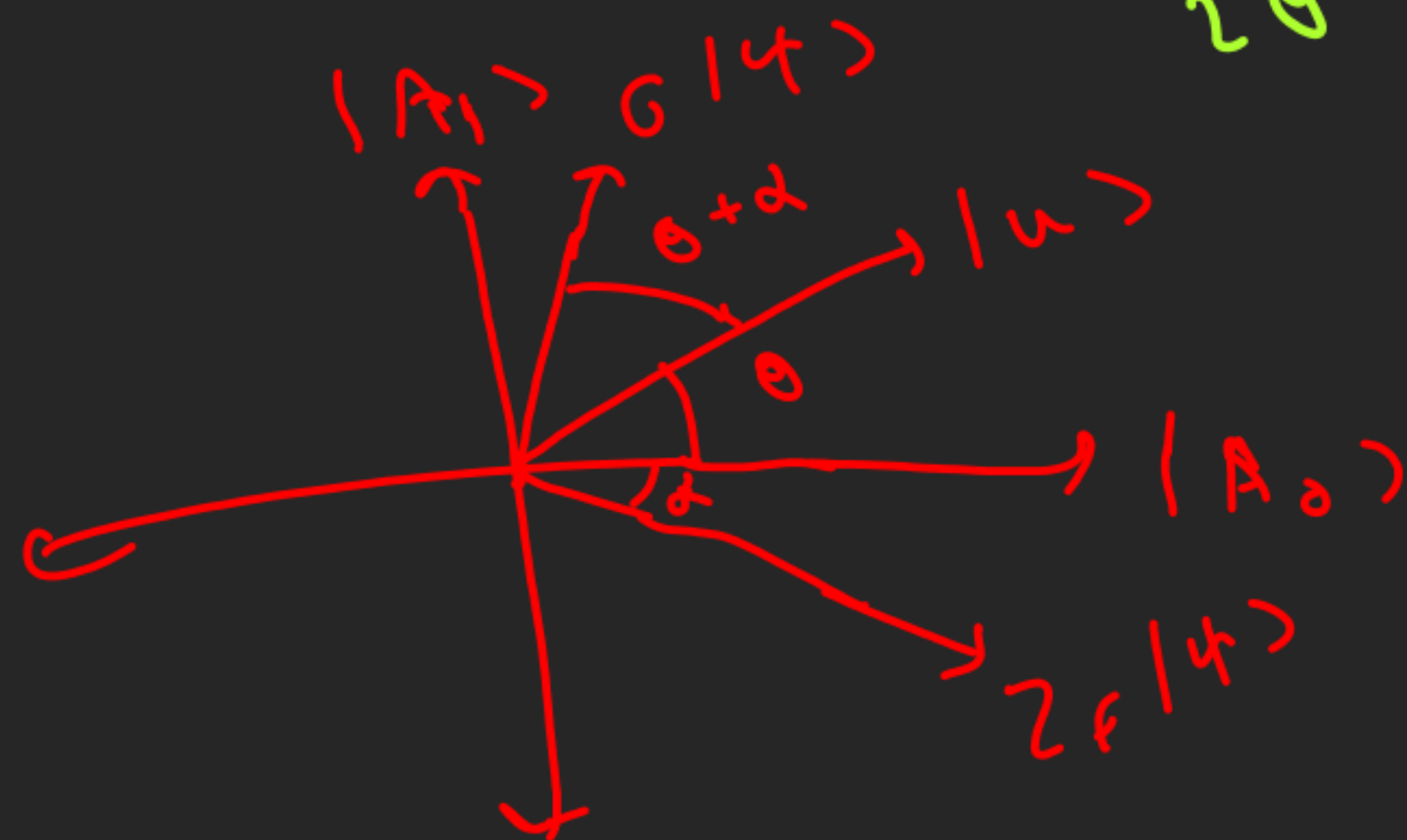


↓
rotated by
 2θ

→ $H^{\otimes n} Z$ or $H^{\otimes n}$

$$(H^{\otimes n} Z \text{ or } H^{\otimes n})|u\rangle = |u\rangle$$

$$(H^{\otimes n} Z \text{ or } H^{\otimes n})|u_{\perp}\rangle = -|u_{\perp}\rangle$$



$$p(N, \underbrace{1}_{|A|}) = \sin^2((2t+1)\theta)$$

N	$p(N, 1)$
2	0.5
4	1
8	0.9453125
16	0.9613190
32	0.9991823

$$p(N, 1) \geq 1 - \frac{1}{N}$$

$$p(N, s) \geq \max \left\{ 1 - \frac{s}{N}, \frac{s}{N} \right\}$$

→ Unknown number of solutions

→ Simple approach

Choose $t \in \left\{1, \dots, \left\lceil \frac{\pi}{4} \sqrt{N} \right\rceil\right\}$ uniformly at random

↳ 40% probability to find a solution

→ Sophisticated Approach

→ Set $T=1$

→ Run Grover's algorithm with $t \in \{1, \dots, T\}$ chosen uniformly at random

